
ONCOSCAN

An AI-Powered Skin Cancer Detection Web Application

1. What is OncoScan?

OncoScan is a smart web application that uses artificial intelligence to help people check moles and skin spots for possible signs of skin cancer. Instead of guessing, the user simply uploads a clear photo of the skin spot (or takes one with their phone or laptop camera). OncoScan studies the image and gives a risk assessment, a possible type of skin spot, and simple educational guidance.

The app always makes it clear that this is not a medical diagnosis. It is an educational tool meant to raise awareness and encourage people to see a doctor when something looks unusual. In this way, OncoScan supports early attention to skin health while keeping the final decision with qualified medical professionals.

2. Why Do We Need This Project?

Skin cancer is one of the most common forms of cancer. Many people notice a new mole or a changing skin spot but are unsure whether it needs attention. Visiting a dermatologist for every small change is not always easy or affordable, and delay can sometimes be harmful.

At the same time, free online tools that claim to “detect cancer” can be misleading or unsafe if they do not explain their limits. There is a need for a simple, honest, and educational system that uses modern AI carefully — showing a risk estimate and confidence, while clearly telling the user to consult a doctor. OncoScan is designed to fill that gap as a student-friendly, responsible awareness project.

3. Main Features (What OncoScan Can Do)

Image Upload & Camera Capture

The user can upload an existing photo or capture a new one using the device camera. The app shows a preview so the user can check that the image is clear and well-lit before analysis.

Skin Spot Classification

A deep-learning model studies the image and suggests a possible category for the skin spot (for example benign, suspicious, or a more specific class depending on the trained model). Results are always presented as suggestions, not final answers.

Risk / Confidence Score

Along with the suggested type, OncoScan shows how confident the model is. This helps the user understand that the system is not 100% sure and that professional advice is still needed.

Educational Guidance

The app provides short, clear tips — such as watching for changes in size, colour, or shape, and when it is sensible to book an appointment with a dermatologist.

Strong Safety Disclaimer

Every result screen includes a clear message: “This is not a medical diagnosis. Please consult a qualified doctor for any skin concern.” This keeps the project honest and responsible.

4. How OncoScan Works (Simple Step-by-Step)

The process is designed to be easy to understand and quick to use:

Step 1 – Capture or Upload the Image

The user opens the web app on a phone or laptop, then either uploads a photo of the skin spot or uses the camera to take a new one. A preview confirms the image is suitable.

Step 2 – Image Processing (AI Analysis)

- The image is cleaned and prepared (resized, normalised, optionally improved for lighting).
- A trained deep-learning model (such as EfficientNet, ResNet, or MobileNet) classifies the skin spot.
- The model also produces a confidence score that shows how sure it is about the result.

Step 3 – Show the Result Clearly

The app displays the suggested type of skin spot, the confidence/risk score, and short educational tips. A strong medical disclaimer appears on the same screen so the user never mistakes the tool for a doctor's opinion.

Step 4 – Simple, Accessible Interface

The web app uses large buttons, clear labels, high contrast, and optional voice output so it is easy to use for a wide range of people.

5. Technology Used

OncoScan is built with modern, widely available technologies so it can run in a normal web browser:

Layer	Technologies
Frontend (what the user sees)	HTML, CSS, JavaScript, React Browser camera & file upload APIs Optional Web Speech API for spoken results
Backend (server side)	Python with Flask or Django REST APIs for image upload and results Optional SQLite for consented scan history
AI / Vision (the “brain”)	OpenCV for image preparation TensorFlow / Keras or PyTorch EfficientNet, ResNet, or MobileNet models HAM10000 or ISIC datasets for training / evaluation

6. Who Does OncoScan Help?

- Anyone who notices a new or changing mole or skin spot and wants a careful first look at home.
- People who need clear, simple guidance on when it is sensible to visit a dermatologist.
- Students and developers who want to learn how AI can support health awareness in a responsible way.
- Communities where access to specialists is limited and early educational tools can encourage timely care.

7. Important Design Principles

Safety First

OncoScan never claims to diagnose cancer. Every result screen carries a clear disclaimer and encourages the user to seek professional medical advice. Language is calm and educational, not alarming.

Privacy-Aware

Images are not stored by default. If the user chooses to keep a history of scans, explicit consent is required and deletion is always possible. Processing is kept as private as practical.

Honest About Limits

Results are approximate. Accuracy can fall with poor lighting, blur, unusual angles, or different skin types. The confidence score and disclaimer help users understand these limits.

Accessibility

The interface uses semantic HTML, keyboard support, screen-reader labels, high contrast, and optional spoken results so a wide range of users can use the tool comfortably.

8. Expected Benefits

- Greater awareness — users learn to pay attention to moles and skin spots earlier.
- Easy first step — a private, at-home check that can prompt a timely doctor visit when needed.
- Clear communication — confidence scores and disclaimers reduce the chance of misunderstanding.
- Educational value — students gain hands-on experience with real AI, image processing, and ethical design.
- Privacy and control — users decide whether any image or history is kept.

9. Conclusion

OncoScan is a practical, AI-powered web application created to help people check moles and skin spots in a careful and responsible way. By combining image analysis, a clear risk estimate, educational tips, and a strong medical disclaimer, it aims to support early awareness without ever replacing a doctor.

The project balances modern deep-learning technology with safety, privacy, and honesty. It does not claim to diagnose skin cancer. Instead, it offers an accessible learning and awareness tool that can encourage users to seek professional care when something looks unusual — and that teaches builders how to create AI systems that respect both people and medical limits.

OncoScan · AI-Powered Skin Spot Awareness · Education · Safety · Privacy